

Reynolds Number and Flow Regimes Homework

Student Version

Fluid Dynamics

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Use SI units unless otherwise stated. For circular-pipe flow, classify the regime using laminar below 2300, transitional from 2300 to 4000, and turbulent above 4000. State the representative velocity and characteristic length.

Homework 1: Water Flow in a Circular Pipe

Problem

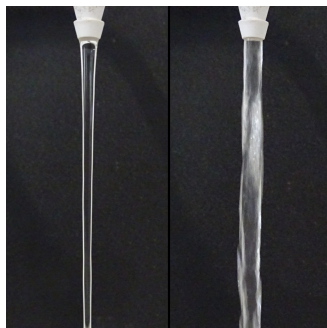
Water at 20°C flows through a circular pipe with diameter $D = 30$ mm at a volume flow rate

$$Q = 1.80 \times 10^{-4} \text{ m}^3/\text{s}.$$

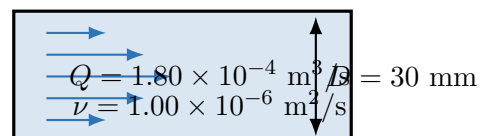
Use $\nu = 1.00 \times 10^{-6} \text{ m}^2/\text{s}$. Determine:

1. the cross-sectional mean velocity;
2. the pipe Reynolds number;
3. the flow regime.

Real flow visualization



Engineering diagram



Circular water pipe: $D = 30$ mm, $Q = 1.80 \times 10^{-4} \text{ m}^3/\text{s}$, $\nu = 1.00 \times 10^{-6} \text{ m}^2/\text{s}$.

Hint

Use

$$A = \frac{\pi D^2}{4}, \quad U = \frac{Q}{A}, \quad \text{Re}_D = \frac{UD}{\nu}.$$

Convert the diameter to meters before calculating.

Solution

Convert the diameter:

$$D = 30 \text{ mm} = 0.030 \text{ m}.$$

The pipe area is

$$A = \frac{\pi(0.030)^2}{4} = 7.069 \times 10^{-4} \text{ m}^2.$$

The mean velocity is

$$U = \frac{Q}{A} = \frac{1.80 \times 10^{-4}}{7.069 \times 10^{-4}} = 0.255 \text{ m/s}.$$

The Reynolds number is

$$\text{Re}_D = \frac{UD}{\nu} = \frac{(0.255)(0.030)}{1.00 \times 10^{-6}} = 7.64 \times 10^3.$$

Since $\text{Re}_D > 4000$, the flow is turbulent.

$$U = 0.255 \text{ m/s}, \quad \text{Re}_D = 7.64 \times 10^3, \quad \text{turbulent}$$

Homework 2: Maximum Laminar Oil Flow Rate

Problem

An oil with kinematic viscosity

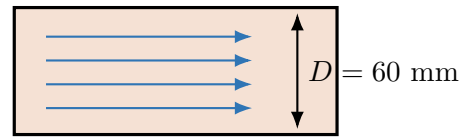
$$\nu = 1.80 \times 10^{-4} \text{ m}^2/\text{s}$$

flows through a 60 mm-diameter circular pipe. Using the conservative laminar limit $\text{Re}_{D,\max} = 2000$, determine:

1. the maximum allowable mean velocity;
2. the maximum volume flow rate in m^3/s ;
3. the maximum volume flow rate in m^3/h .

Real piping application**Engineering diagram**

$$\text{Re}_{D,\text{max}} = 2000$$



$$\nu = 1.80 \times 10^{-4} \text{ m}^2/\text{s}$$

Oil pipe: $D = 60 \text{ mm}$, $\nu = 1.80 \times 10^{-4} \text{ m}^2/\text{s}$, conservative limit $\text{Re}_D = 2000$.

Hint

Rearrange

$$\text{Re}_D = \frac{UD}{\nu}$$

to obtain U_{max} . Then use

$$Q_{\text{max}} = U_{\text{max}} \frac{\pi D^2}{4}.$$

Multiply m^3/s by 3600 to obtain m^3/h .

Solution

Convert the diameter:

$$D = 0.060 \text{ m}.$$

The maximum mean velocity is

$$U_{\text{max}} = \frac{\text{Re}_{D,\text{max}} \nu}{D} = \frac{(2000)(1.80 \times 10^{-4})}{0.060} = 6.00 \text{ m/s}.$$

The pipe area is

$$A = \frac{\pi(0.060)^2}{4} = 2.827 \times 10^{-3} \text{ m}^2.$$

Therefore,

$$Q_{\text{max}} = U_{\text{max}} A = (6.00)(2.827 \times 10^{-3}) = 1.696 \times 10^{-2} \text{ m}^3/\text{s}.$$

Converting to cubic meters per hour,

$$Q_{\text{max}} = (1.696 \times 10^{-2})(3600) = 61.1 \text{ m}^3/\text{h}.$$

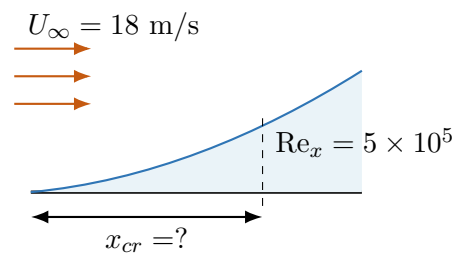
$$U_{\text{max}} = 6.00 \text{ m/s}, \quad Q_{\text{max}} = 1.70 \times 10^{-2} \text{ m}^3/\text{s} = 61.1 \text{ m}^3/\text{h}$$

Homework 3: Transition Location on a Flat Plate**Problem**

Air flows over a smooth flat plate with free-stream velocity $U_\infty = 18$ m/s. Use

$$\nu = 1.50 \times 10^{-5} \text{ m}^2/\text{s}$$

and estimate the distance from the leading edge at which $\text{Re}_x = 5.0 \times 10^5$.

Real transition visualization**Boundary-layer diagram**

Flat plate: $U_\infty = 18$ m/s, $\nu = 1.50 \times 10^{-5} \text{ m}^2/\text{s}$, $\text{Re}_{x,cr} = 5.0 \times 10^5$.

Hint

Use

$$\text{Re}_x = \frac{U_\infty x}{\nu}$$

and solve for x . Remember that this is an estimate and not a universal transition location.

Homework 4: Wind-Tunnel Reynolds-Number Similarity

Problem

A 0.45 m-long model car is tested in air at a wind-tunnel speed of 30 m/s. Use

$$\nu_{air} = 1.50 \times 10^{-5} \text{ m}^2/\text{s}.$$

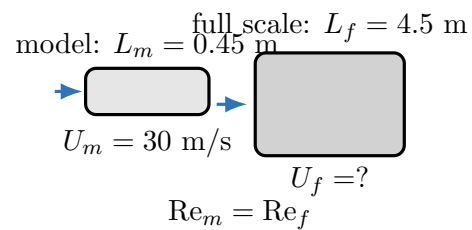
Determine:

1. the model Reynolds number based on car length;
2. the speed required for a geometrically similar 4.5 m-long full-scale car in the same air to have the same Reynolds number.

Real wind tunnel



Similarity diagram



Same air: model $L_m = 0.45 \text{ m}$, $U_m = 30 \text{ m/s}$; full scale $L_f = 4.5 \text{ m}$.

Hint

For the model,

$$Re_m = \frac{U_m L_m}{\nu}.$$

For equal Reynolds numbers in the same fluid,

$$U_m L_m = U_f L_f.$$

Homework 5: Effect of Water Temperature on Reynolds Number

Problem

Water flows through a 20 mm-diameter pipe at a mean velocity of 0.40 m/s. Compare the Reynolds number and flow regime at two temperatures:

$$\nu_{20^\circ\text{C}} = 1.00 \times 10^{-6} \text{ m}^2/\text{s}, \quad \nu_{60^\circ\text{C}} = 0.475 \times 10^{-6} \text{ m}^2/\text{s}.$$

Determine:

1. Re_D at 20°C ;
2. Re_D at 60°C ;
3. the regime at each temperature;
4. why heating the water changes Reynolds number even though the pipe and mean velocity are unchanged.

Real piping application



Property-comparison diagram

$$D = 0.020 \text{ m}, U = 0.40 \text{ m/s}$$



$$20^\circ\text{C}: \nu = 1.00 \times 10^{-6} \text{ m}^2/\text{s}$$



$$60^\circ\text{C}: \nu = 0.475 \times 10^{-6} \text{ m}^2/\text{s}$$

$$\nu \downarrow \Rightarrow \text{Re} \uparrow$$

Same pipe and speed at both temperatures; only kinematic viscosity changes.

Hint

Use

$$\text{Re}_D = \frac{UD}{\nu}$$

for each temperature. A lower kinematic viscosity produces a larger Reynolds number when U and D are fixed.

End of Reynolds Number and Flow Regimes Homework